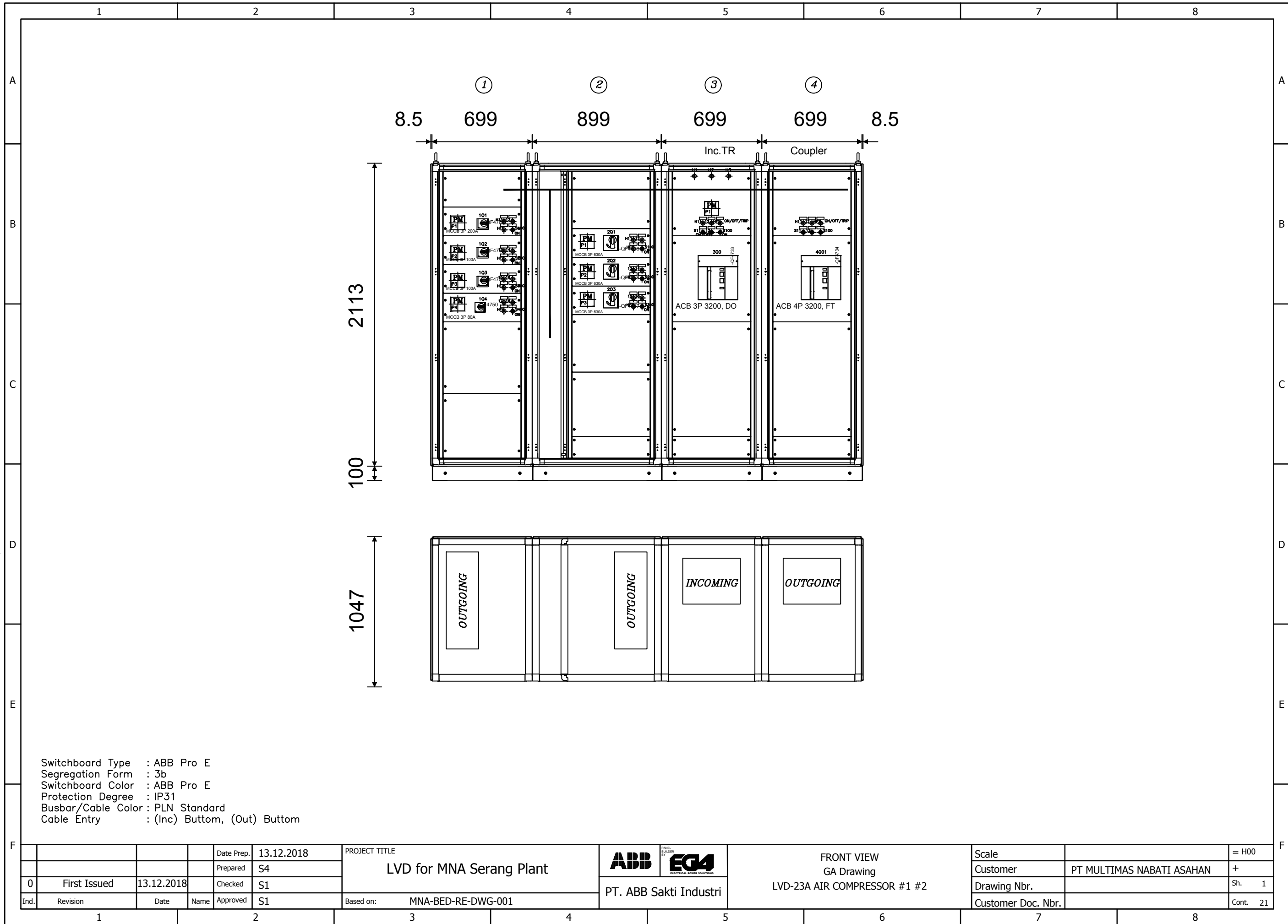
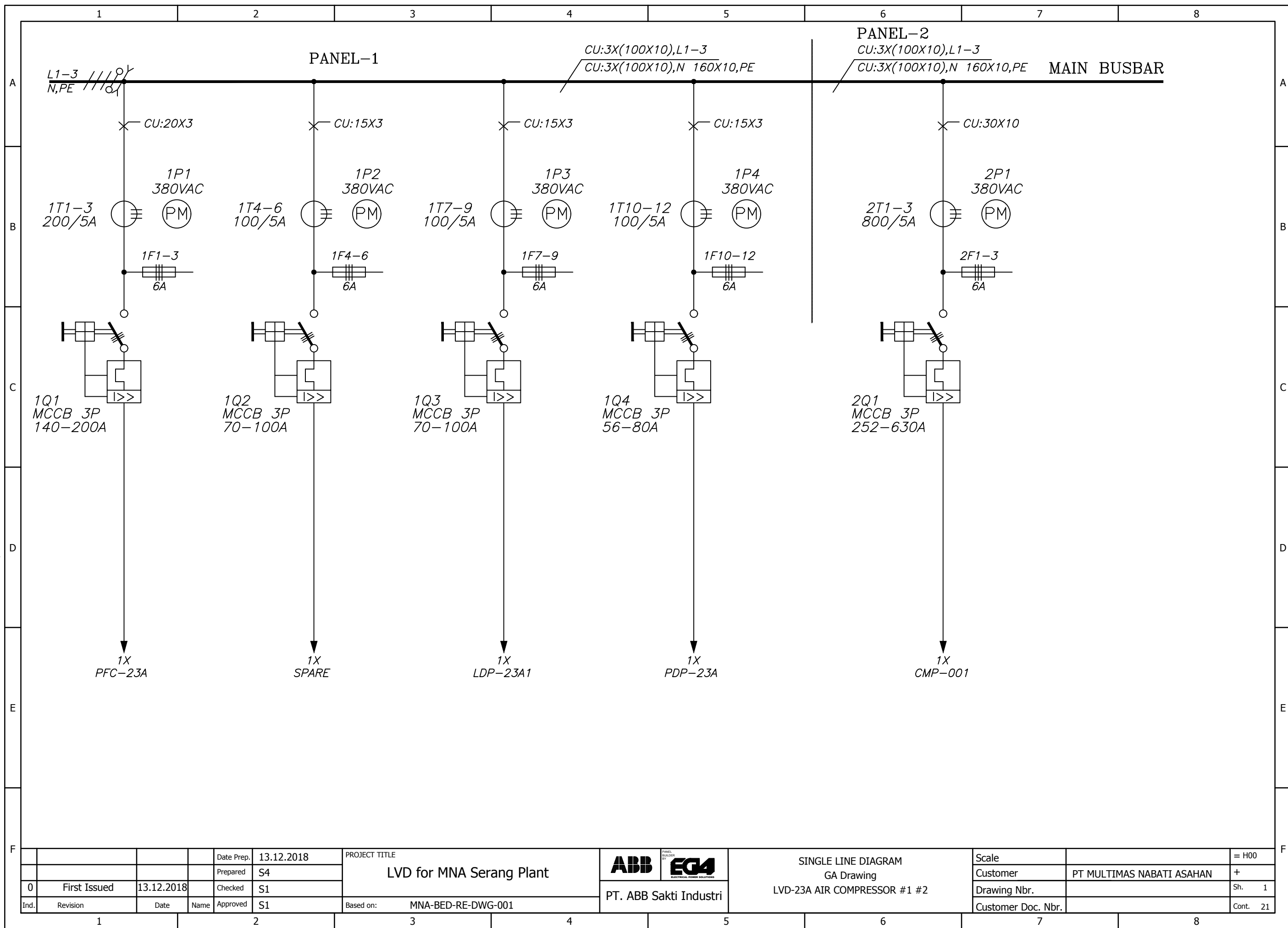


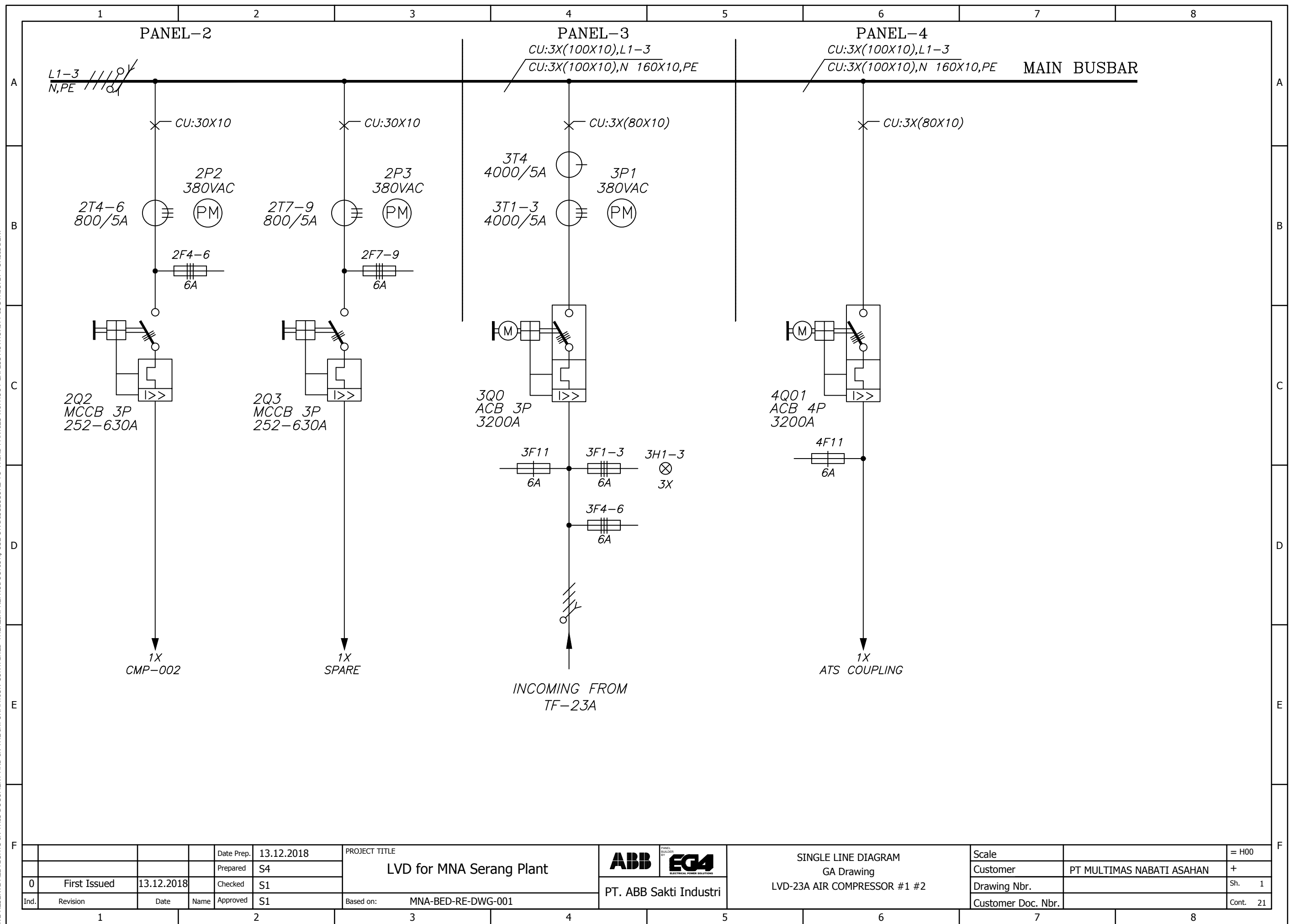
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

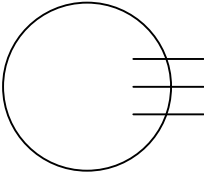
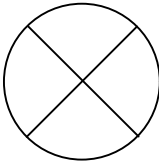
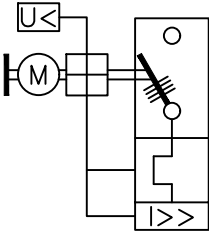
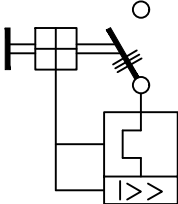


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

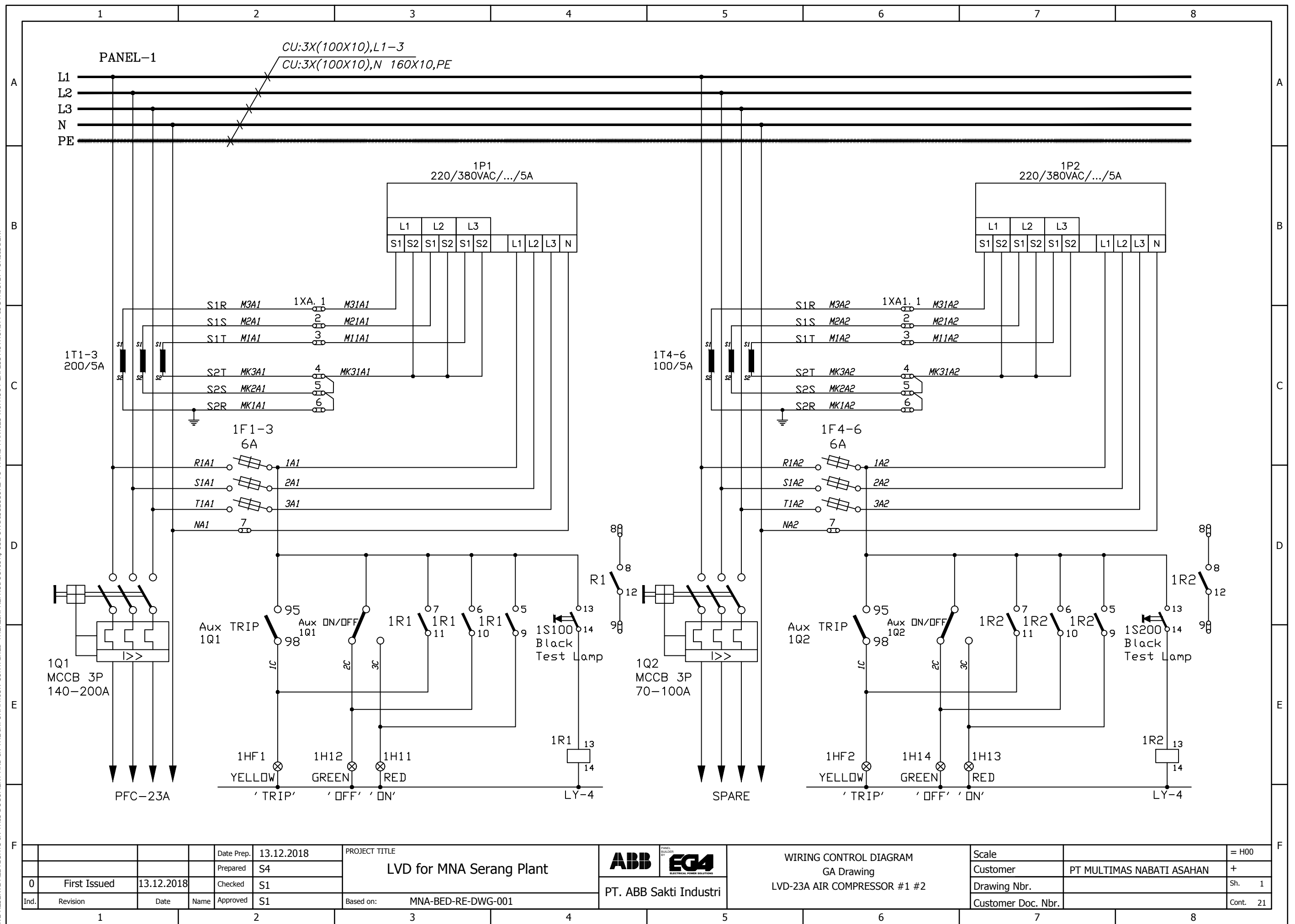


				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant			SINGLE LINE DIAGRAM GA Drawing LVD-23A AIR COMPRESSOR #1 #2	Scale		= H00	
				Prepared	S4					Customer	PT MULTIMAS NABATI ASAHAN	+	
0	First Issued	13.12.2018		Checked	S1		PT. ABB Sakti Industri	Drawing Nbr.			Sh.	1	
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001				Customer Doc. Nbr.		Cont.	21

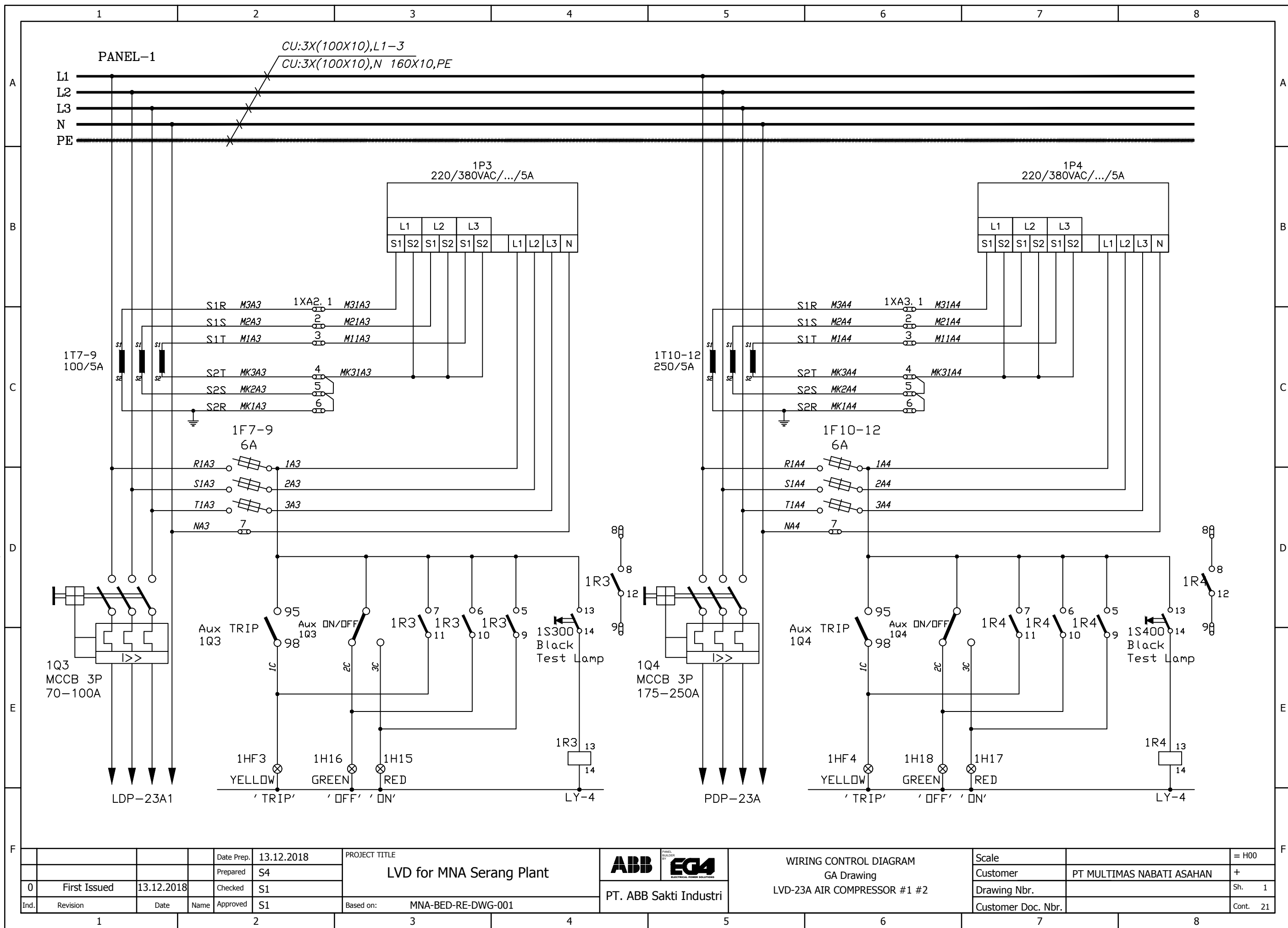
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

	1	2	3	4	5	6	7	8				
A												
B												
C		CT							PILOT LAMP			
D		ACB							FUSE			
E												
F		MCCB										
		DIGITAL POWER METER										
0	First Issued	13.12.2018		Checked	S1	LEGEND GA Drawing LVD-23A AIR COMPRESSOR #1 #2			Scale		= H00	
	Revision	Date	Name	Approved	S1				PT. ABB Sakti Industri	Customer	PT MULTIMAS NABATI ASAHAN	+
										Drawing Nbr.		Sh.
									Customer Doc. Nbr.		Cont.	21

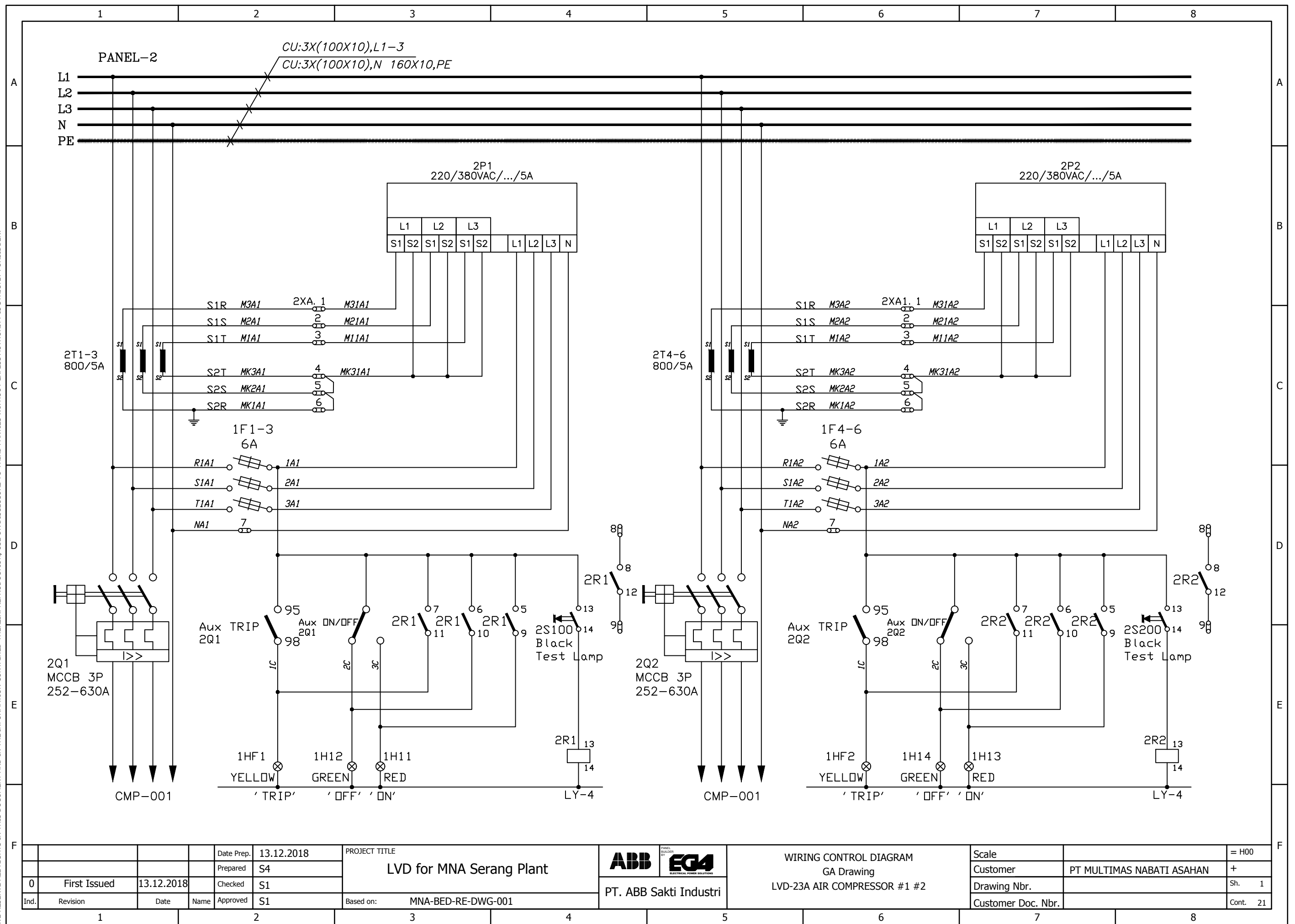
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



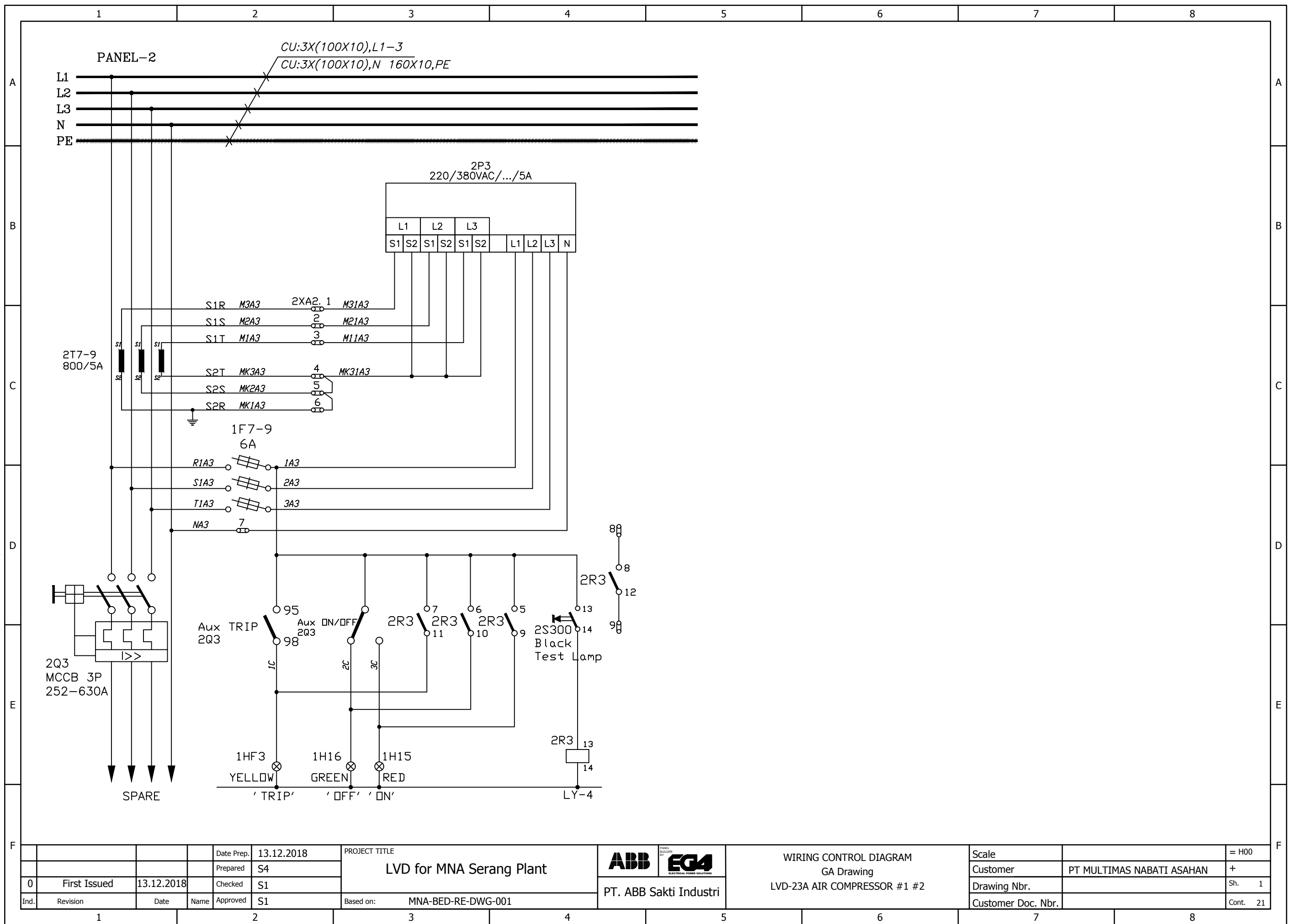
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



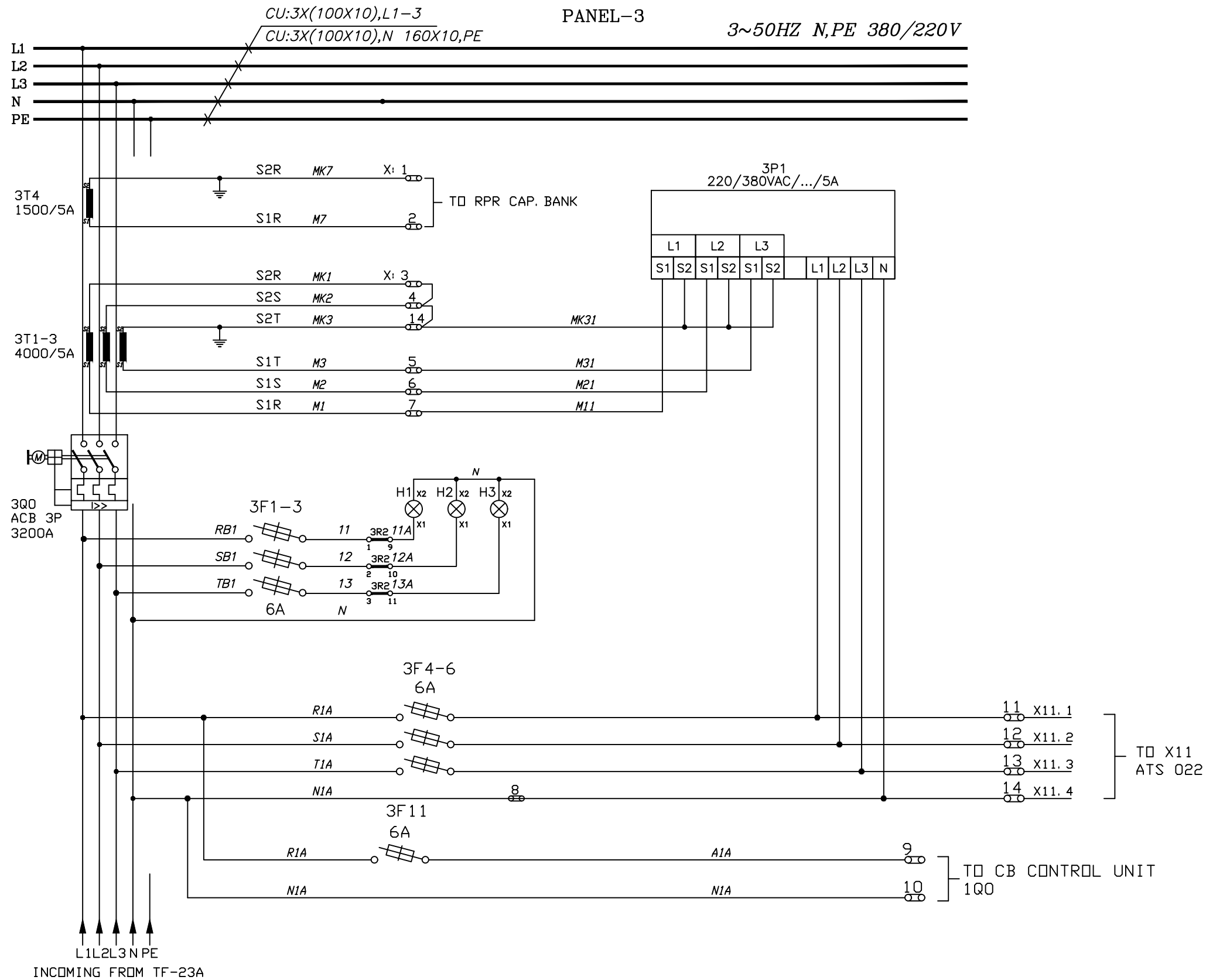
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

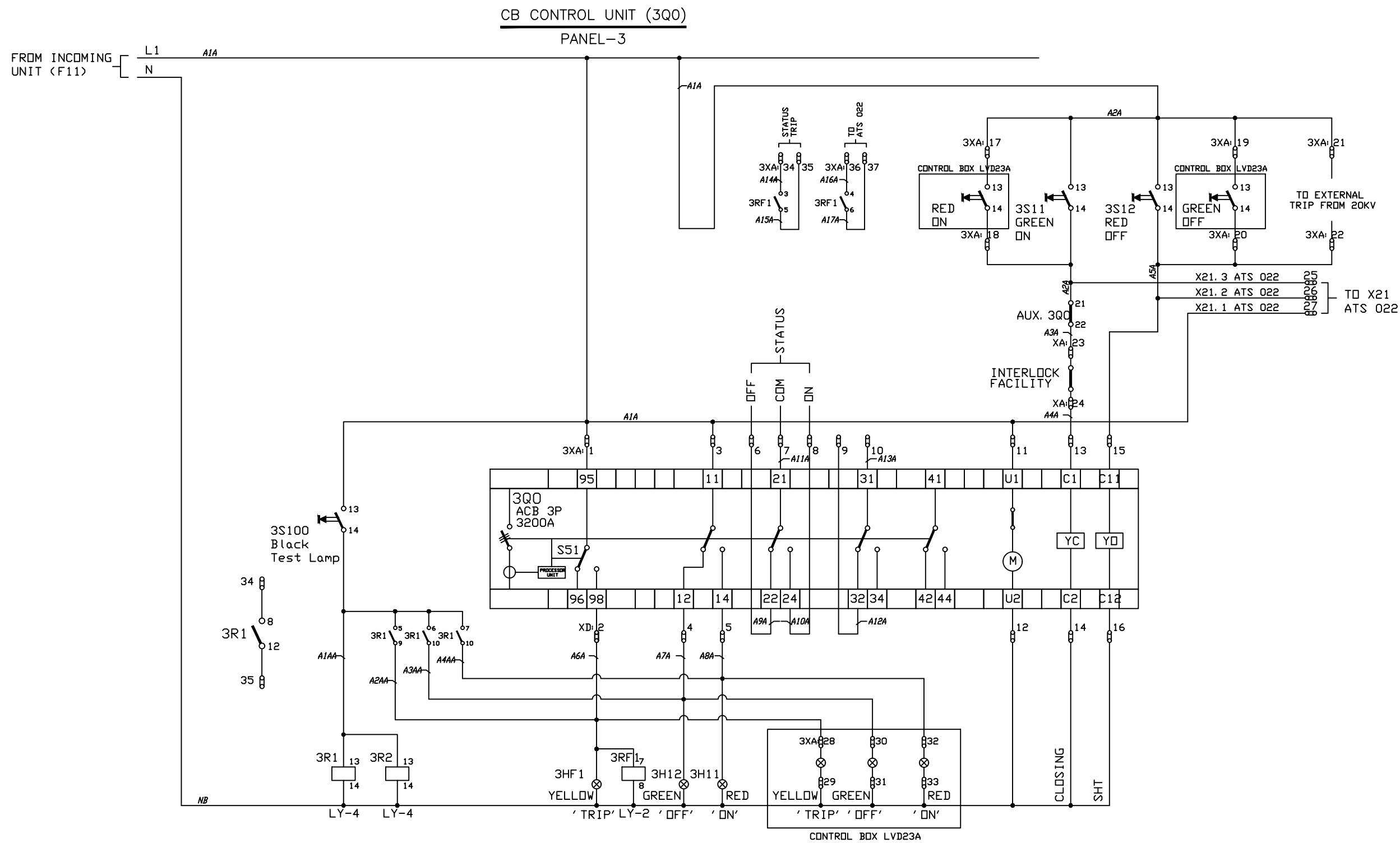


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



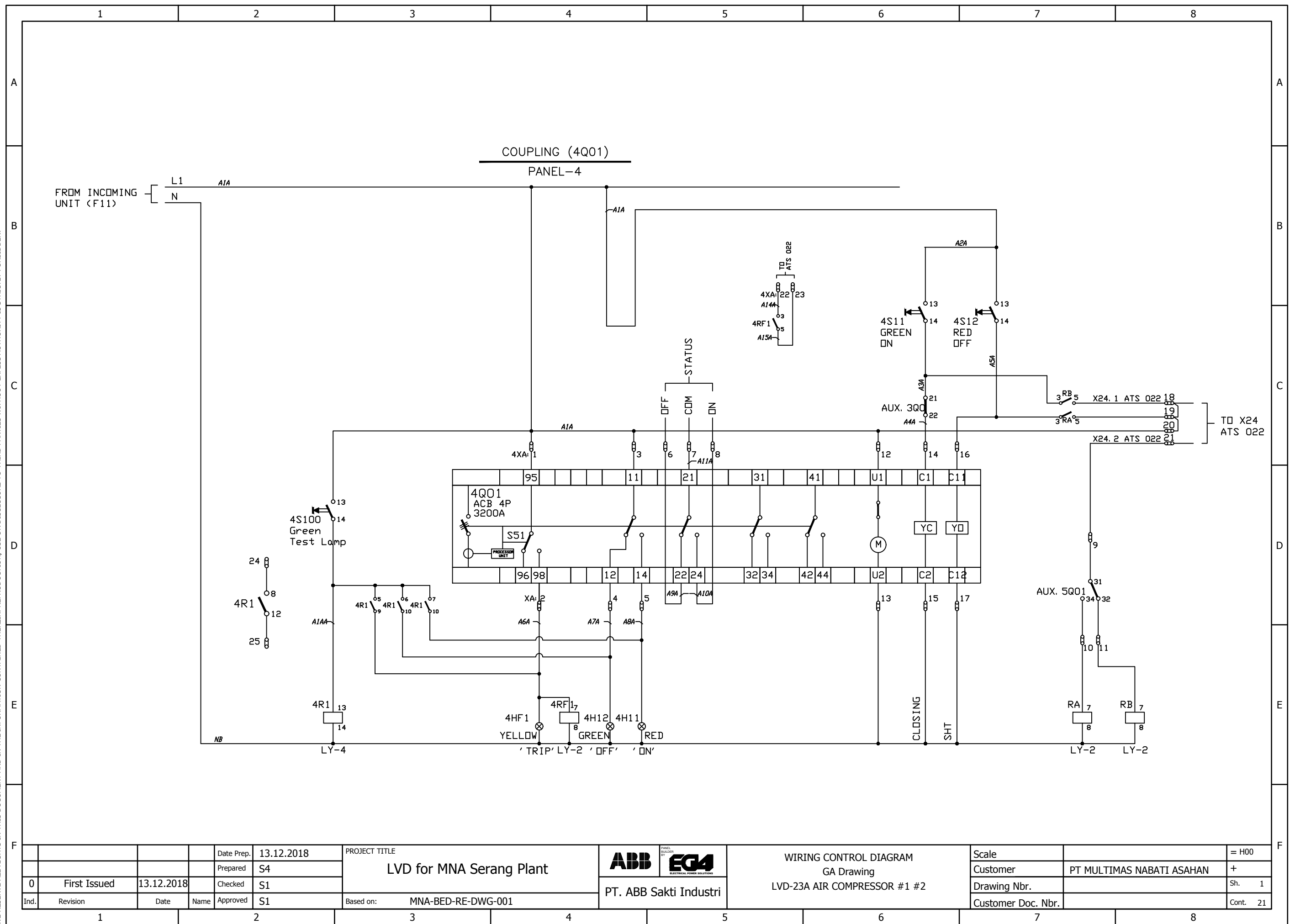
		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		  PT. ABB Sakti Industri	WIRING CONTROL DIAGRAM GA Drawing LVD-23A AIR COMPRESSOR #1 #2	Scale		= H00
		Prepared	S4					Customer	PT MULTIMAS NABATI ASAHAN	+
0		First Issued	13.12.2018					Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.		Cont. 21

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

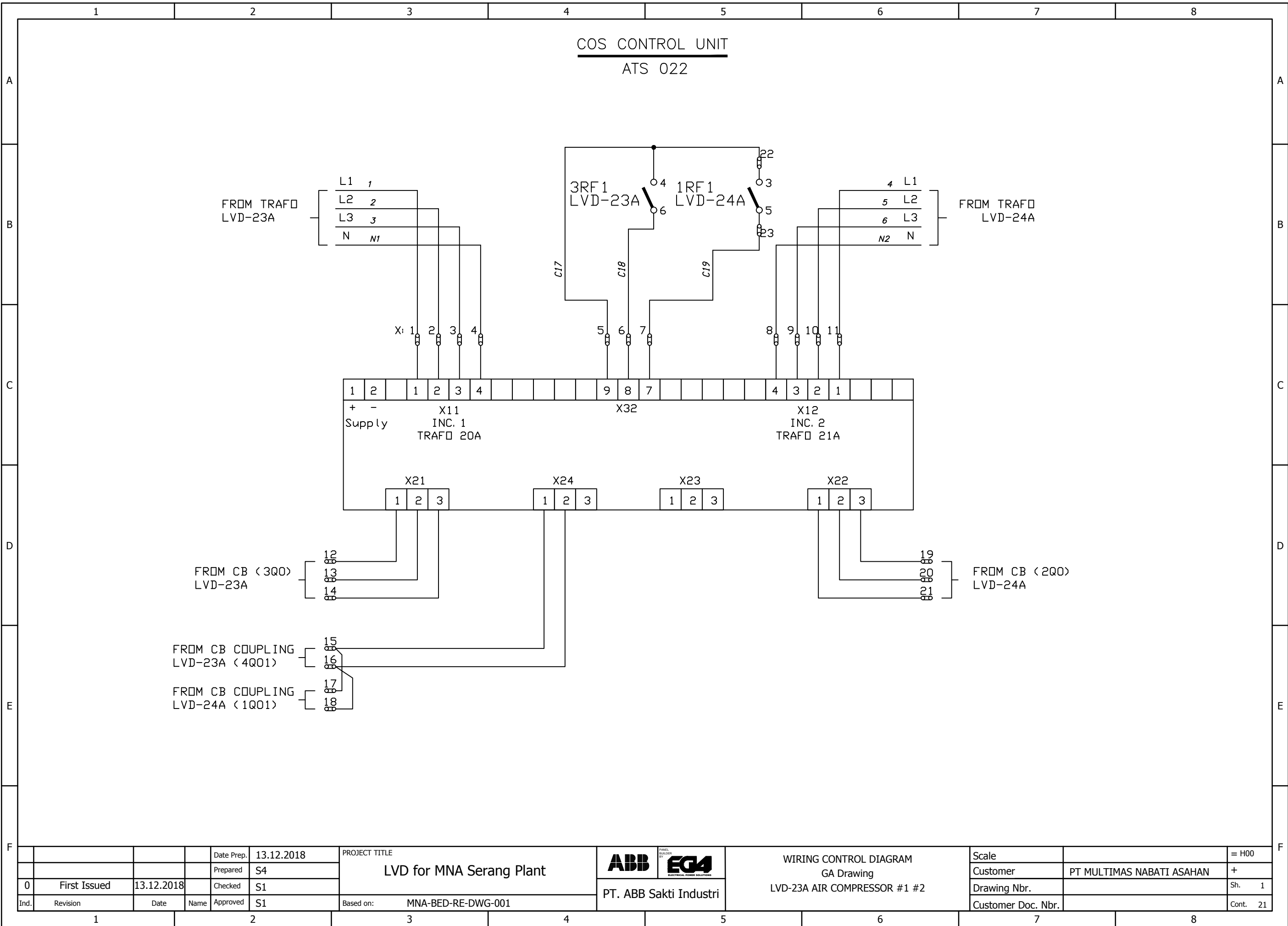


			Date Prep.	13.12.2018	PROJECT TITLE	 	WIRING CONTROL DIAGRAM GA Drawing LVD-23A AIR COMPRESSOR #1 #2	Scale		= H00
			Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018	Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 21

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8			
A	N O	DESCRIPTION			TECHNICAL DATA			QTY	DESIGNATION						A		
		INCOMING															
B	1	Fuse Carrier 6A			AXLE/Setara			7	3F11,1F1-7						B		
	2	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE/Setara			2	3H1,11								
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE/Setara			2	3H2,HF1								
	4	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE/Setara			2	3H3,12								
	5	DIGITAL POWER METER (*)			ABB, M2M ETHERNET			1	3P1								
	6	ATS 022 (*)			ABB, ATS 022			1	-								
	7	CT 4000/5A CL.0.5 40VA			Howig, HG 120-4000			4	3T1-4								
C	8	ACB 3P 3200A, 66KA, DO, Ekip Touch LSIG (*)			ABB, E4.2N32 D/O			1	3Q0						C		
	9	+ Shunt Trip Release (Y0) (*)			ABB, Y0			1	-								
	10	+ Motor (M) + Closing Release (YC) For E2.2-E6.2 (*)			ABB, M+YC			1	-								
	11	+ Padlocks For E2.2-E6.2 (*)			ABB,			1	-								
	12	+ Ekip Com Protocol IEC 61850 (*)			ABB,			1	-								
	13	Control Relay 220Vac			Omron, LY-2			1	RF1								
	14	Control Relay 220Vac			Omron, LY-4			2	R1-2								
D	15	Push Button On/Test Lamp , 1NO, Black			Ega			1	S100						D		
	16	Push Button On, 1NO, Green			Ega			1	3S11								
	17	Push Button Off, 1NO, Red			Ega			1	3S12								
	18																
	19	COUPLER															
	20	Fuse Carrier 6A			AXLE/Setara			1	4F8								
	21	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE/Setara			1	4H11								
E	22	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE/Setara			1	4HF1						E		
	23	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE/Setara			1	4H12								
	24	ACB 4P 3200A, 66KA, FT, Ekip Dip LI (*)			ABB, E4.2N32 F/T			1	4Q01								
	25	Push Button On/Test Lamp , 1NO, Black			Ega			1	S100								
	26	Control Relay 220Vac			Omron, LY-4			1	R1								
	F																F
					Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant			  PT. ABB Sakti Industri		COMPONENT LIST GA Drawing LVD-23A AIR COMPRESSOR #1 #2		Scale		= H00	
					Prepared	S4								Customer	PT MULTIMAS NABATI ASAHAN	+	
0		First Issued	13.12.2018		Checked	S1								Drawing Nbr.		Sh.	
Ind.		Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001					Customer Doc. Nbr.		Cont.	21	
1		2		3		4		5		6		7		8			

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8	
A	N O		DESCRIPTION		TECHNICAL DATA		QTY		DESIGNATION						A
			OUTGOING												
B	1	Fuse Carrier 6A		AXLE		21		1F1-12,2F1-F9						B	
	2	Pilot Lamp (LED) 220Vac/50Hz, Red		AXLE		7		1H11,13,15,17 ; 2H11,13,15							
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow		AXLE		7		1HF1-4 ; 2HF1-3							
	4	Pilot Lamp (LED) 220Vac/50Hz, Green		AXLE		7		1H12,14,16,18 ; 2H12,14,16							
	5	DIGITAL POWER METER (*)		ABB, M2M ETHERNET		7		1P1-4 ; 2P1-4							
	6	CT 800/5A CL.1 5VA		Howig, HG 50I-800		9		2T1-9							
	7	CT 200/5A CL.1 5VA		Howig, HG 30I-200		3		1T1-3							
C	8	CT 100/5A CL.1 2,5VA		Howig, HG 30I-100		9		1T4-12						C	
	9	MCCB 3P 252-630A, 50KA (*)		ABB, T5S PR221DS-LS/I-630		3		2Q1-3							
	10	+ Auxiliary Contact For T5-T6 (*)		ABB, Aux-C 3Q 1SY		3		-							
	11	+ Extended Rotary Handles For T5 (*)		ABB, RHE		3		-							
	12	MCCB 3P 140-200A, 50KA (*)		ABB, XT4S TMA 200		1		1Q1							
	13	+ Auxiliary Contact For XT2/XT4 (*)		ABB, Aux-C 2Q 1SY		1		-							
	14	+ Extended Rotary Handles For XT2/XT4 (*)		ABB, RHE		1		-							
D	15	MCCB 3P 70-100A, 50KA (*)		ABB, XT2S TMA-100		2		1Q2-Q3						D	
	16	+ Auxiliary Contact For XT2/XT4 (*)		ABB, Aux-C 2Q 1SY		2		-							
	17	+ Extended Rotary Handles For XT2/XT4 (*)		ABB, RHE		2		-							
	18	MCCB 3P 56-80A, 50KA (*)		ABB, XT2S TMA-80		1		1Q4							
	19	+ Auxiliary Contact For XT2/XT4 (*)		ABB, Aux-C 2Q 1SY		1		-							
	20	+ Extended Rotary Handles For XT2/XT4 (*)		ABB, RHE		1		-							
	21	Control Relay 220Vac		Omron, LY-4		7		1R1-4 ;2R1-3							
E	22	Push Button Test Lamp , 1NO, Black		Ega		7		1S100-400 ; 2S100-300						E	
	23														
	24														
	25														
	26														
	26														
F															F
1		2		3		4		5		6		7		8	